



Designing the Connection



“If 10 computers are connected in a lab,
what connection pattern should we use?”



What is Network Topology?



Network topology means the arrangement or layout of computers, cables, and network devices in a computer network.

Simple Definition

Topology = how nodes and links are connected.



Key Terms



Node

Any device in a network: computer, printer, server, switch, router.

Link

The connection between two nodes. It may be wired or wireless.

Data Packet

A small unit of data that travels from sender to receiver.



Key Terms



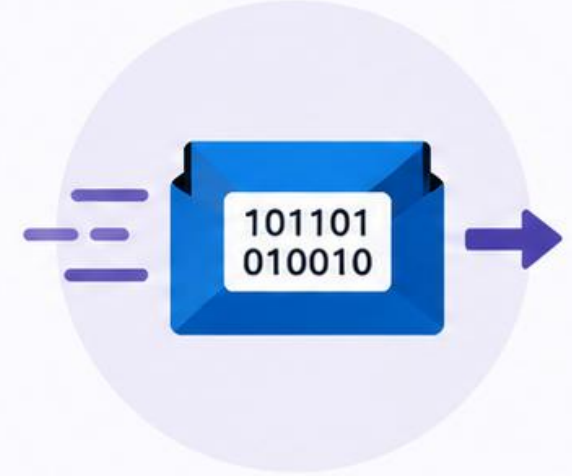
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Link

The connection between two nodes. It may be wired or wireless.



Data Packet

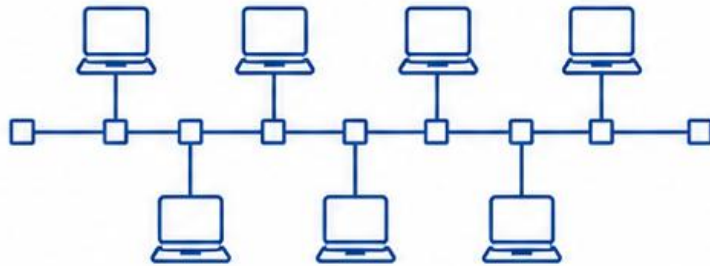
A small unit of data that travels from sender to receiver.



Types of Network Topology

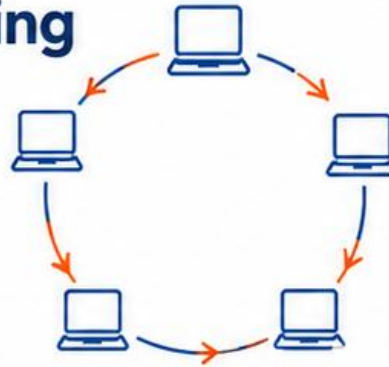


1 Bus



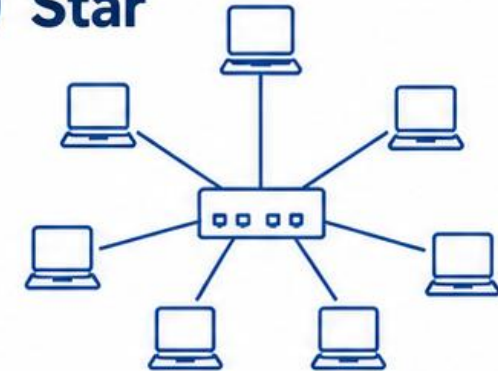
All devices connect to one main cable.

2 Ring



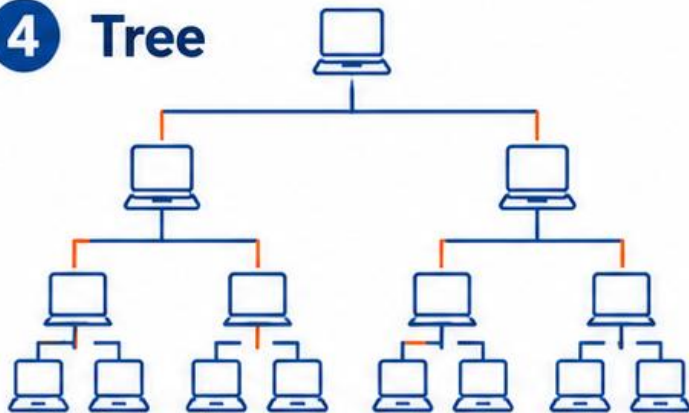
Each device connects in a circular loop.

3 Star



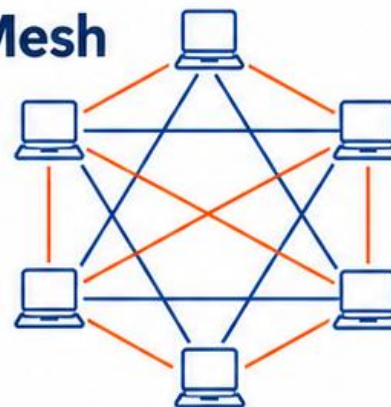
All devices connect to a central hub or switch.

4 Tree



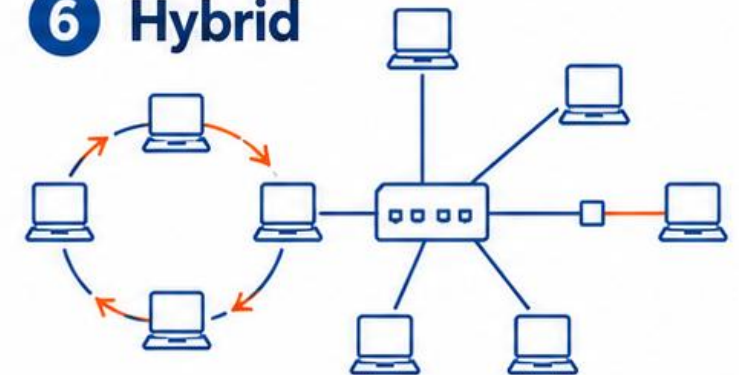
Devices connect in levels like branches of a tree.

5 Mesh



Devices connect to many or all other devices.

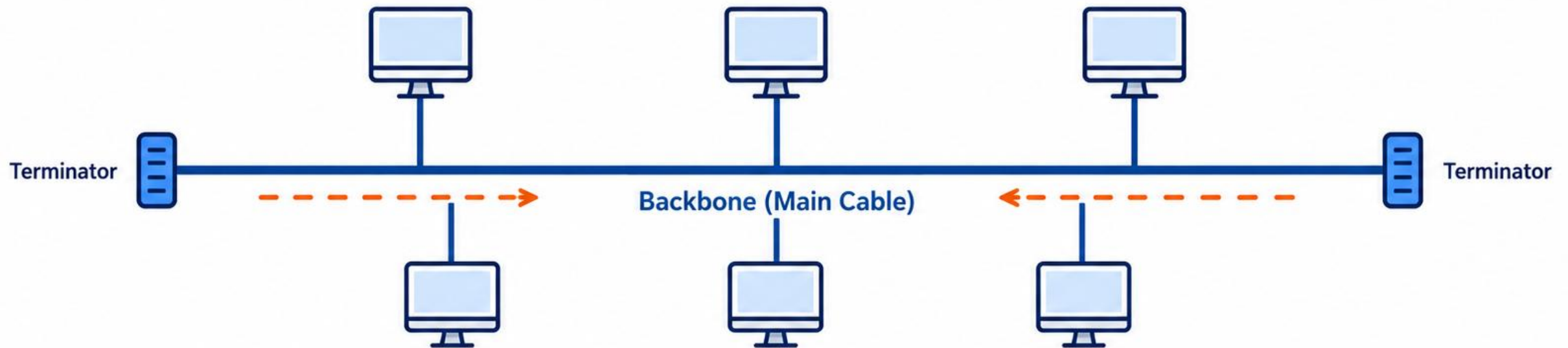
6 Hybrid



A combination of two or more topology types.



Bus Topology: One Main Cable





Bus Topology: Advantages and Limitations



Advantages

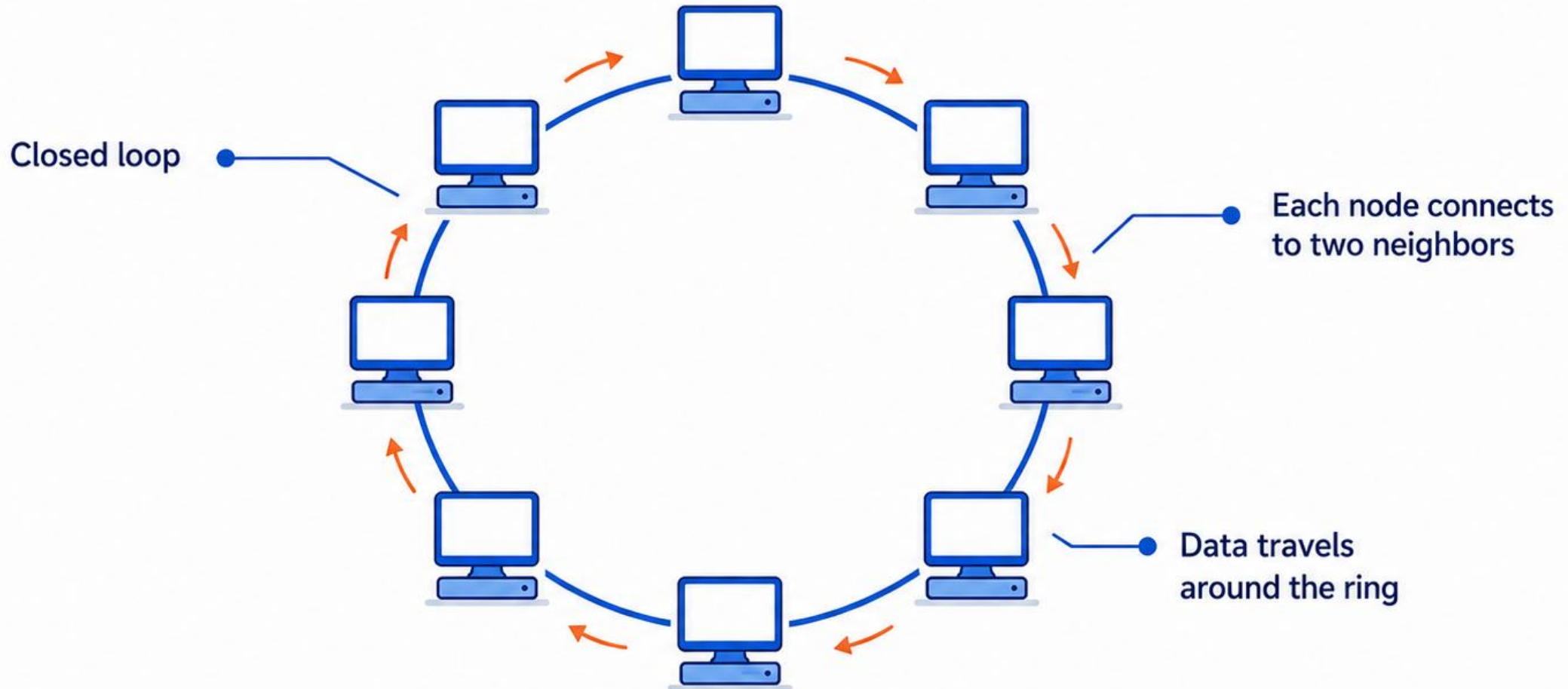
- ☐ Easy to set up.
- ☐ Cheap; needs less cable.
- ☐ Good for small networks.
- ☐ New devices can be added without interruption
- ☐ One computer failure may not stop all.
- ☐ No Centralized Server

Limitations

- ☐ Main cable damage can stop network.
- ☐ Slower with many computers.
- ☐ Higher chance of data collision.
- ☐ Traffic increases with more nodes.



Ring Topology: Computers in a Circle



In a ring topology, devices are connected in a circle, and each device passes data to the next device.



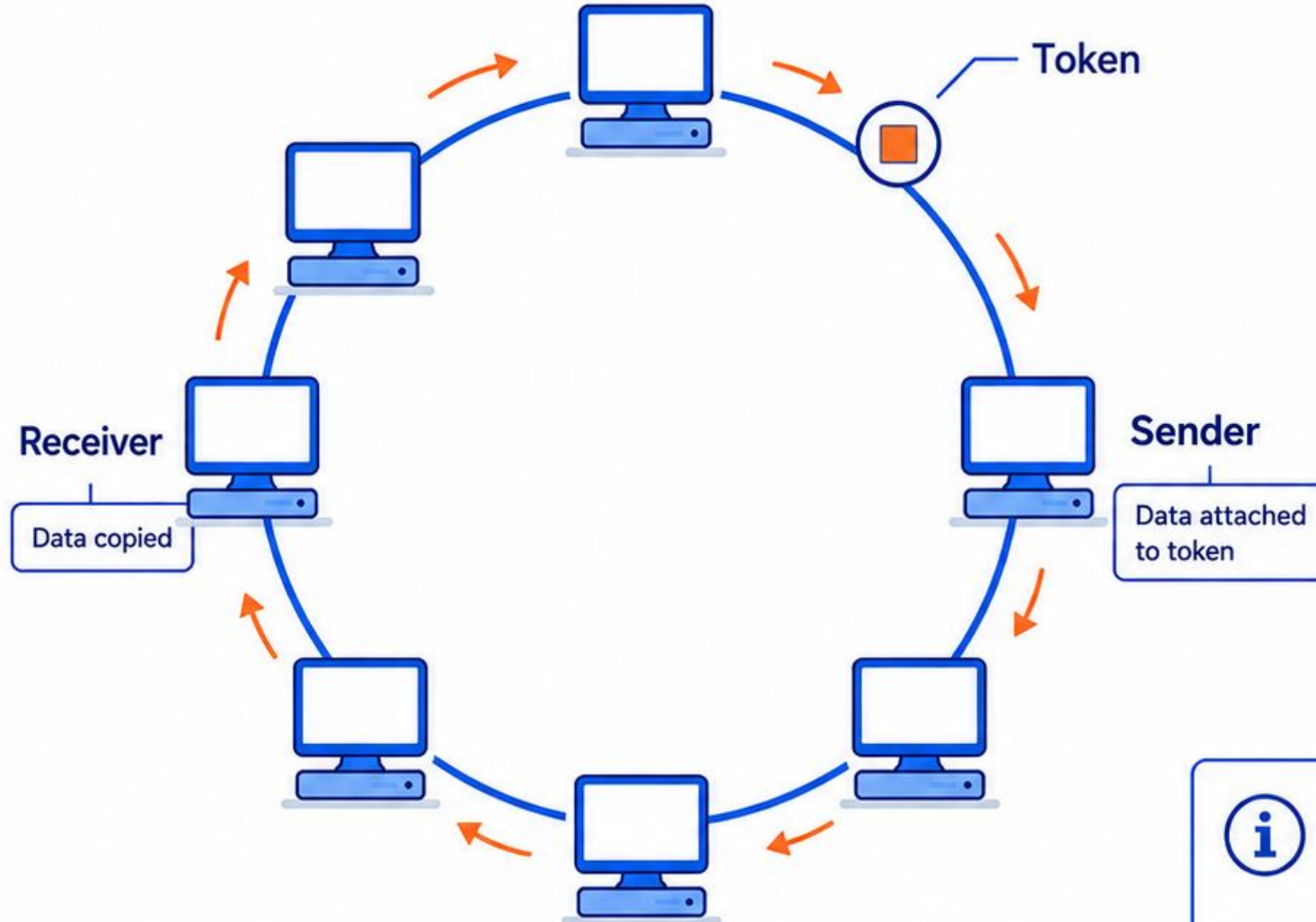
Ring Topology: How to Pass Token



1 Free token circulates around the ring.

2 Sender captures the token and sends data.

3 Receiver copies the data, and the token continues.



i Unidirectional: data moves in one direction only.



Ring Topology: Advantages and Limitations



Advantages

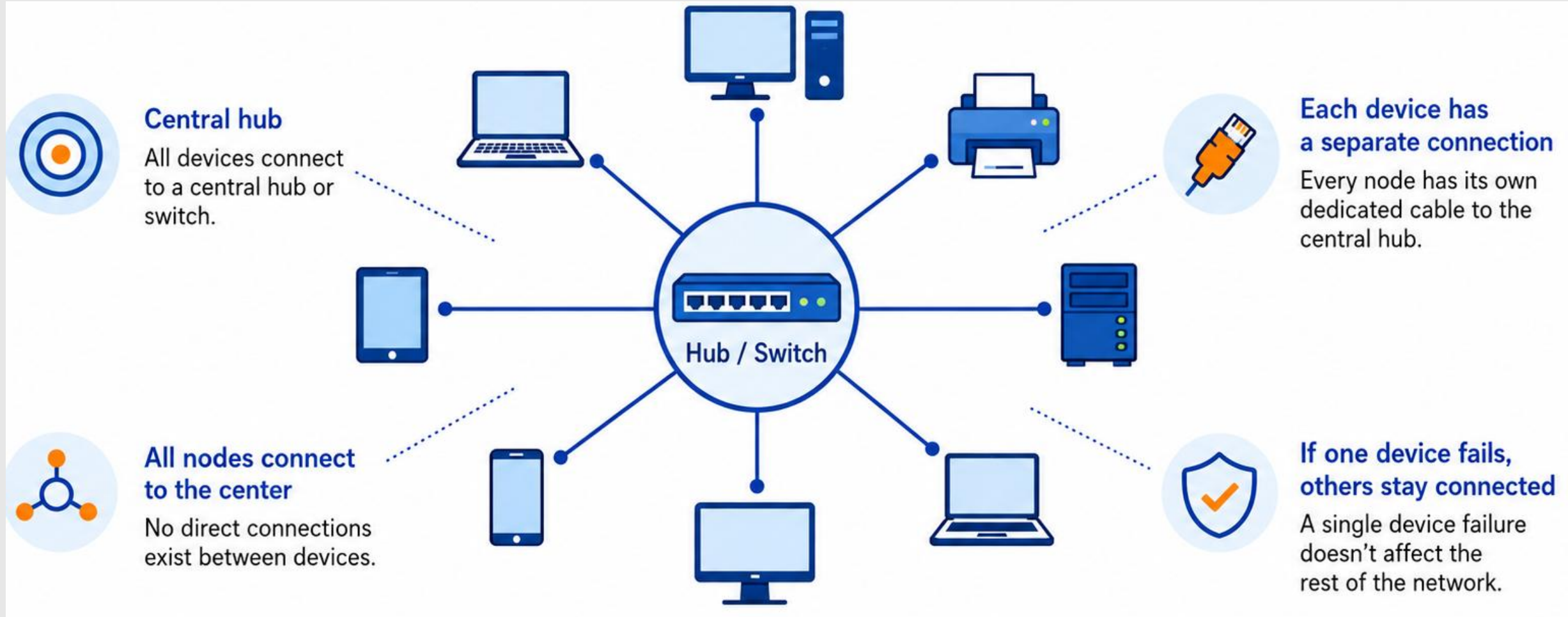
- ☐ Equal chance to send data.
- ☐ No central server required.
- ☐ Less chance of collision.
- ☐ Needs comparatively less cable.

Limitations

- ☐ One device/connection failure may affect all.
- ☐ Adding/removing nodes disrupts network.
- ☐ Error finding can be difficult.
- ☐ More nodes increase transmission time.

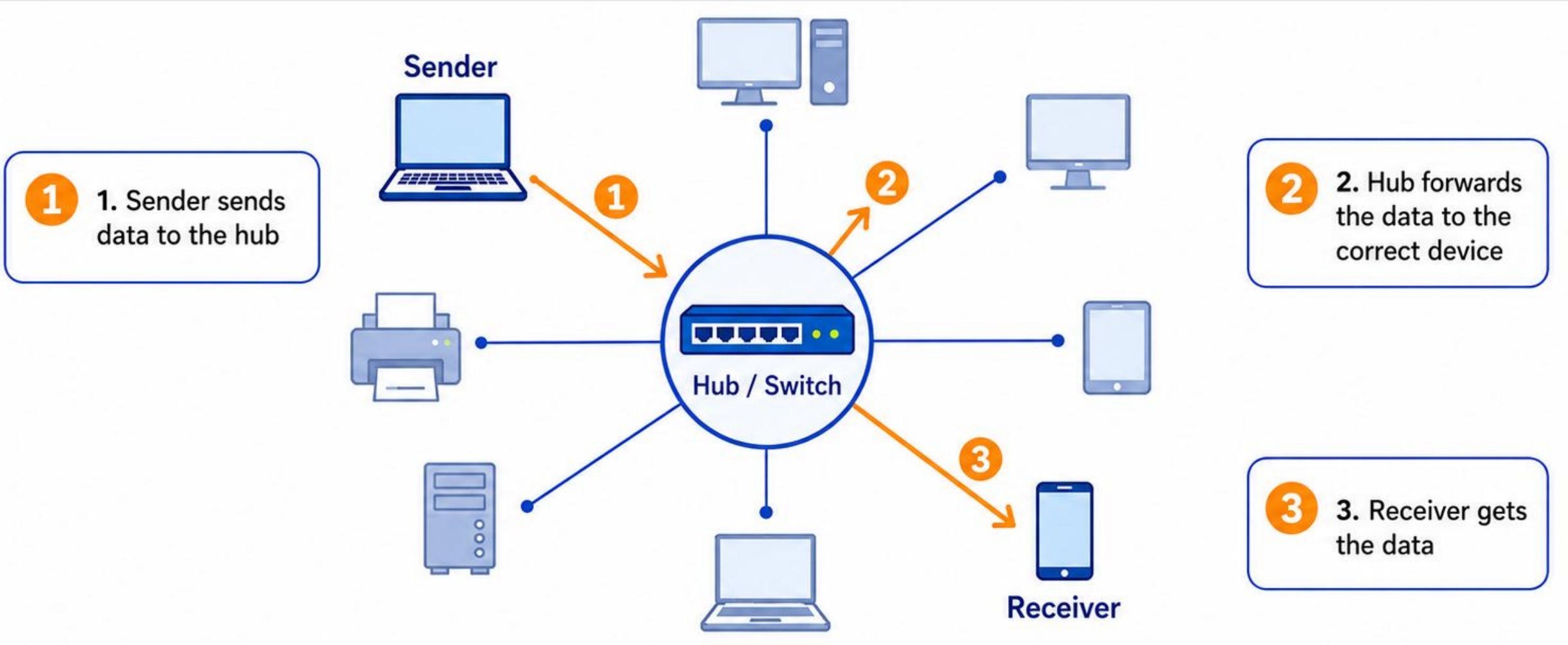


Star Topology: Central Switch/Hub





Star Topology: Central Switch/Hub





Star Topology: Advantages and Limitations



Advantages

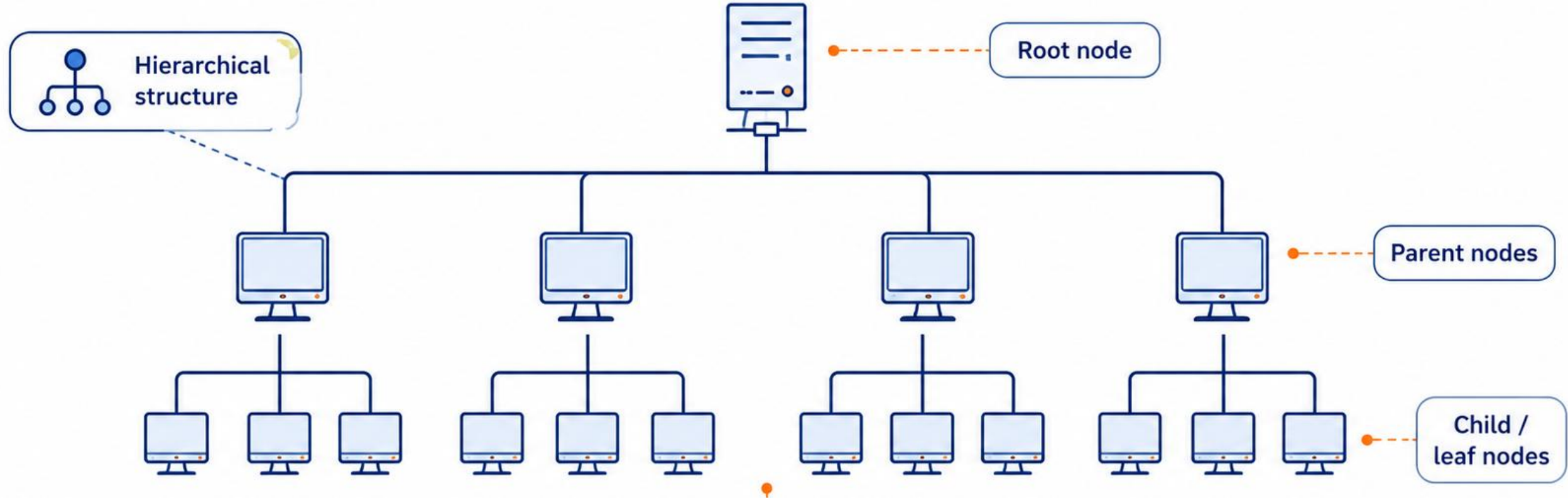
- ☐ Easy to add/remove computers.
- ☐ One PC failure does not stop all.
- ☐ Easy troubleshooting.
- ☐ Faster with a switch.
- ☐ Less chance of collision.

Limitations

- ☐ Central hub/switch failure stops network.
- ☐ Needs more cable.
- ☐ Cost is higher than bus.
- ☐ Devices depend on central device.



Tree Topology: Branch-like Network





Tree Topology: Branch-like Network



Step 1

Data starts at the root or upper-level node



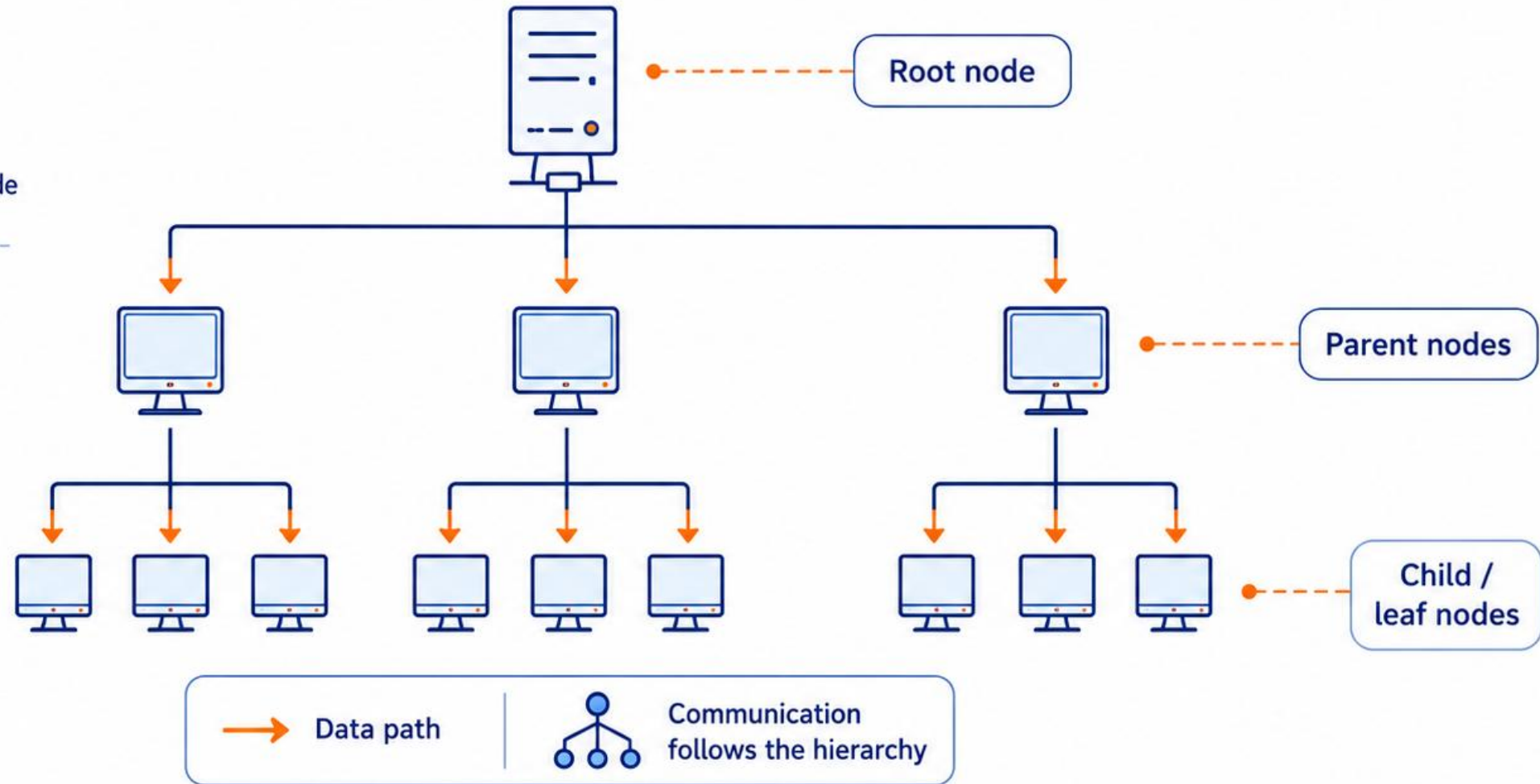
Step 2

Intermediate nodes forward data to the correct branch



Step 3

End devices receive the data at the lower level





Tree Topology: Advantages and Limitations



Advantages

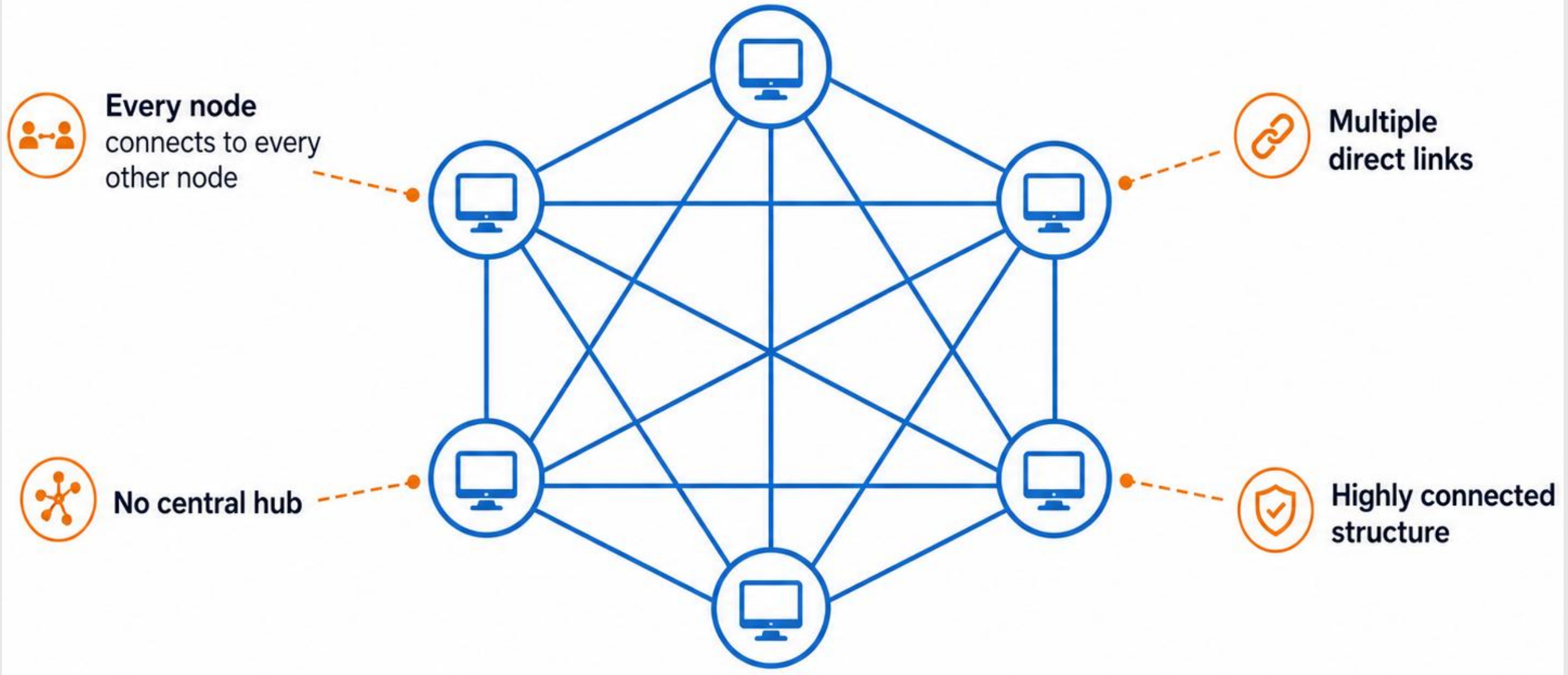
- ☐ Good for large networks.
- ☐ Easy to expand with new branches.
- ☐ Suitable for offices/institutions.
- ☐ If one branch fails, others may work.
- ☐ Organized network management.

Limitations

- ☐ Root device failure can affect all.
- ☐ More complex than star/bus.
- ☐ Needs more cable and devices.
- ☐ Higher cost.
- ☐ Lower-level devices may depend on upper level.



Mesh Topology: Direct Connections





Mesh Topology: Direct Connections



1

1. Sender sends data

The sender transmits data to the network.

2

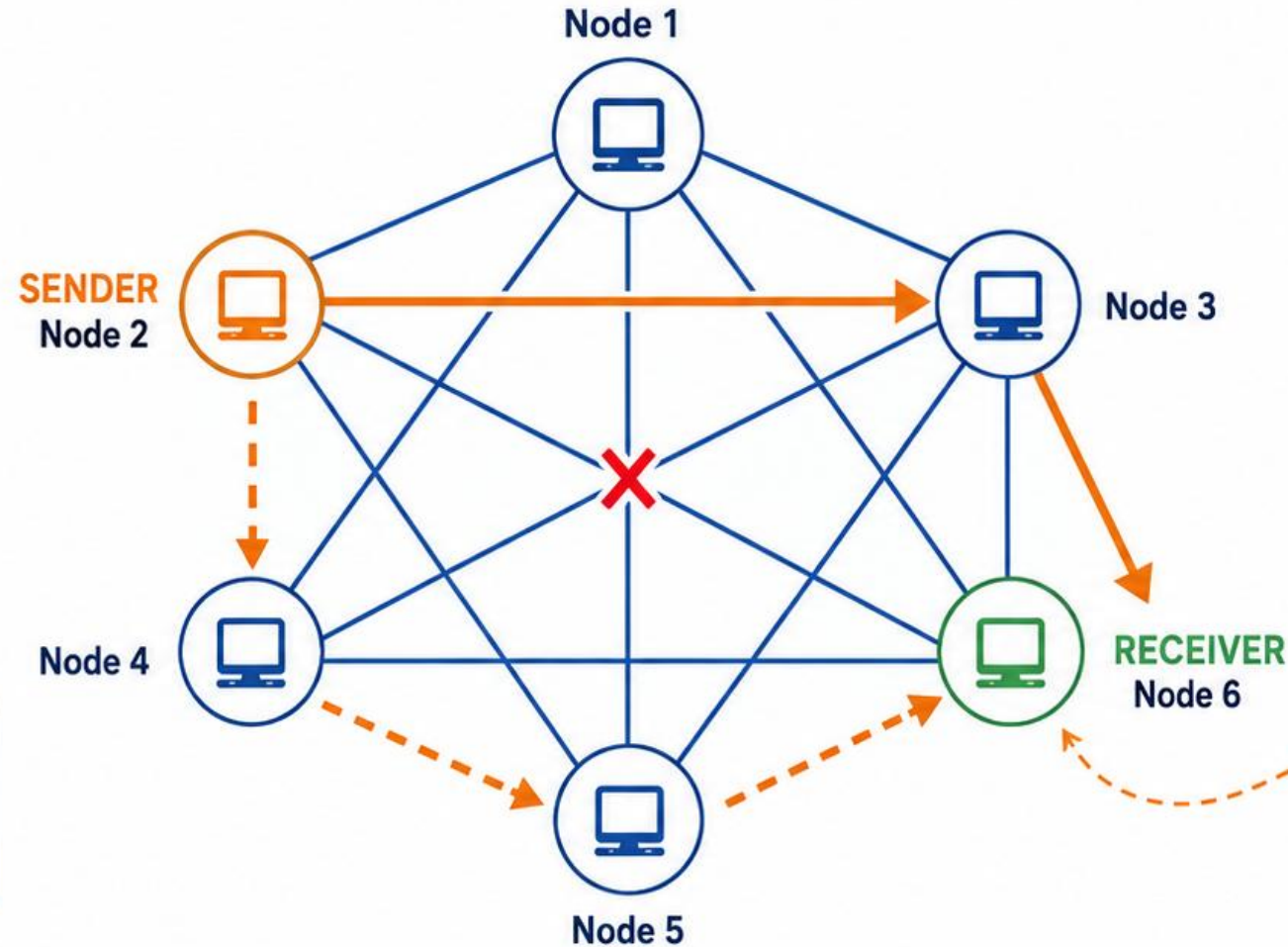
2. Data chooses an available path

Data is routed through one of the available paths.

3

3. Receiver gets the data

The data arrives safely at the receiver.



Backup route available

If one link fails, data can use another path to reach the destination.



In a mesh topology, data can still reach the destination even if one link fails.



Mesh Topology: Connection Formula



For full mesh topology:

$$\text{Connections} = n(n - 1) / 2$$

Task

If there are 6 computers:

$$6(6 - 1) / 2 = 15 \text{ connections}$$



Mesh Topology: Advantages & Limitations



Advantages

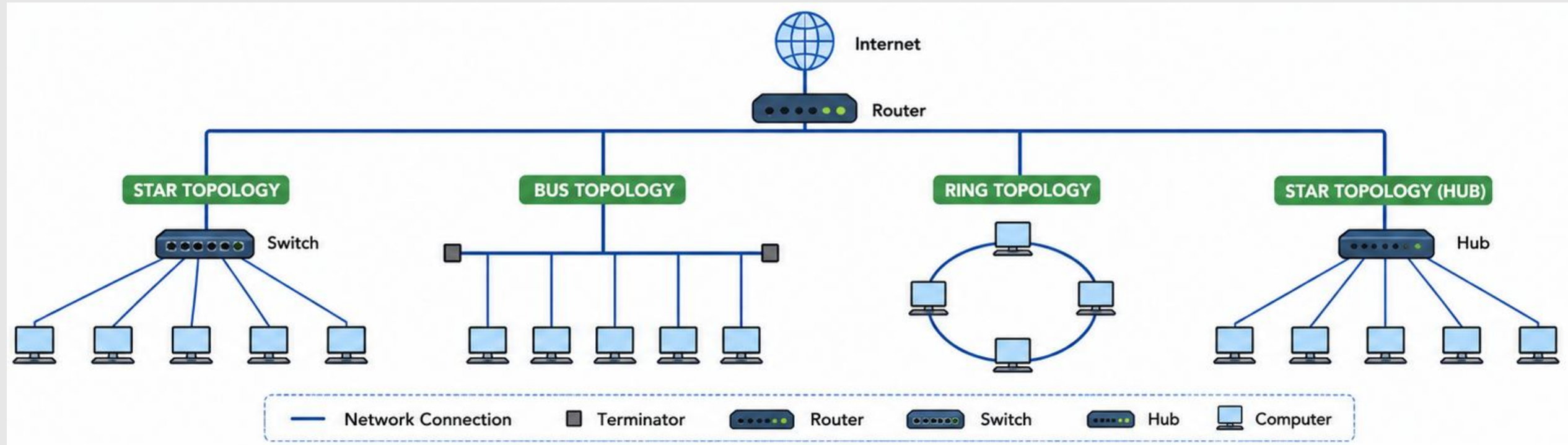
- ☐ Very reliable.
- ☐ Fast communication.
- ☐ Works even if one path fails.
- ☐ Better security.
- ☐ No central server required.

Limitations

- ☐ Very expensive.
- ☐ Needs many cables and ports.
- ☐ Installation is complex.
- ☐ Maintenance is difficult.
- ☐ Cost increases quickly with nodes.



Hybrid Topology: Combination Design





Hybrid Topology: Advantages & Limitations



Advantages

- ☐ Flexible design.
- ☐ Easy to expand.
- ☐ Suitable for large organizations.
- ☐ Failure in one part may not stop all.
- ☐ Combines benefits of topologies.

Limitations

- ☐ Complex design.
- ☐ High maintenance cost.
- ☐ Needs more devices.
- ☐ Requires skilled administrator.
- ☐ Troubleshooting may need planning.



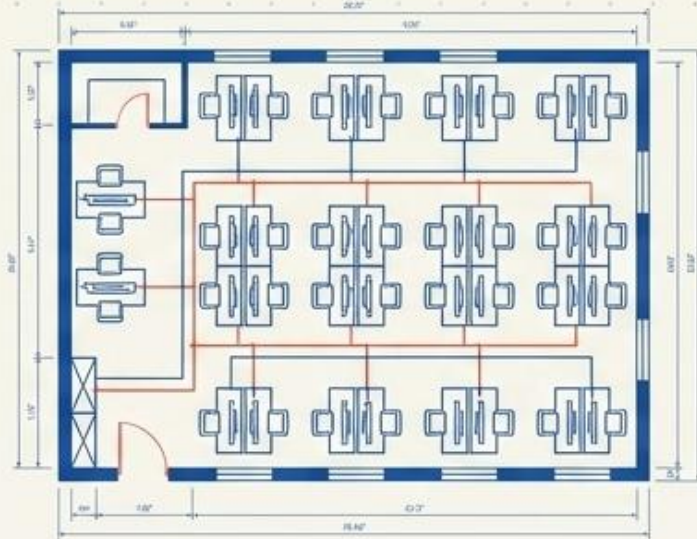
Different Topologies at a Glance



Topology	Structure	Cost	Expansion	Main Issue
Bus	One main cable	Low	Medium	Backbone failure
Ring	Circular path	Medium	Difficult	One break affects all
Star	Central switch	Medium	Easy	Switch failure
Tree	Hierarchy	High	Easy	Root failure
Mesh	All-to-all	Very high	Difficult	Too many links
Hybrid	Combination	High	Easy	Complex maintenance



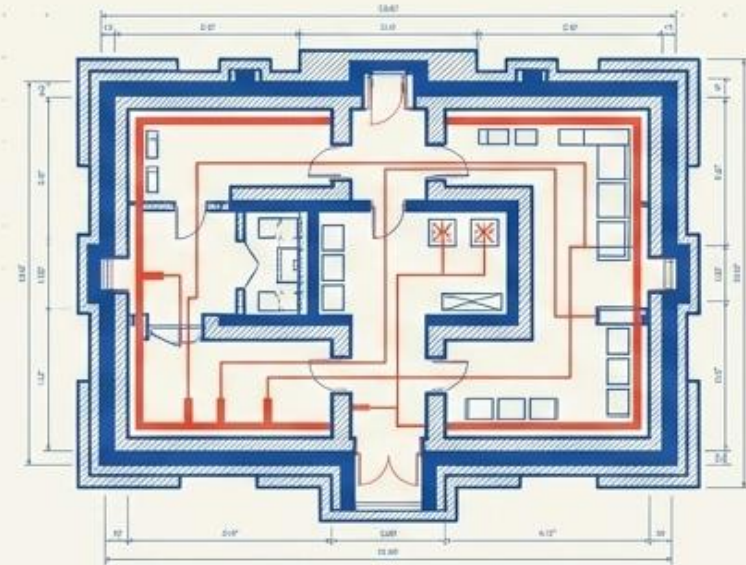
Topology Selection



Scenario A: The Cyber Café

You have 25 computers, a strict budget, and need an easy setup where one broken PC won't ruin the day.

Which topology do you build?



Scenario B: The Defense Base

You have critical data. Cost is not an issue. The network must survive even if multiple cables are physically severed.

Which topology do you build?